MEA OCPP 1.6 Compliance Test Report

Section 2: Auto Charge Verification

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

2026-06-11 12:07

Contents

1 Test Configuration

Device Under Test	Vector vSECC.single Board
Device IP	192.168.1.166
OCPP Version	1.6 (JSON)
CP ID	rddQC4000001
CSMS	wss://ocpp.measandbox.com:2930
Connection	Direct (no proxy)
Test PC	192.168.111.185 (alias 192.168.1.200/24 on enp3s0)
EV Wait Timeout	0 s per plug event
Test Date	2026-06-11 04:24:44 UTC
Results file	tex/vsecc_section2_results.json

Test Architecture

The test runs directly against the MEA CSMS with no proxy. The test PC:

- Calls the vSECC REST API (`http://192.168.1.166/api`) for configuration and status
- Subscribes to the vSECC MQTT broker (`mqtt://192.168.1.166:1883`) to observe StatusNotification, Authorize, Start/StopTransaction, and MeterValues events
- Calls the MEA sandbox REST API (`https://ocppapi.measandbox.com/EV/`) with HTTP Digest authentication for ChangeConfiguration, RemoteStart, and RemoteStop

Items 2.2, 2.5, and 2.14 require a physical EV to be connected to the vSECC connector. If no EV is connected within the configured timeout, those items and all items that depend on an active charging session are recorded as **WARN**.

2 Section 2 Results

Summary: 3 PASS 0 FAIL 18 WARN 0 SKIP (21 total)

Item	Test Description	Detail / Evidence	Result
2.1	ChangeConfiguration AutoCharge=False → Accepted	`HTTP 200`	**PASS**
2.2	StatusNotification Preparing (AutoCharge=False)	`EV_WAIT_SEC=0 | skipped` *Connect physical EV to connector and rerun*	**WARN**
2.3	StatusNotification Available (unplug)	`Depends on 2.2 (EV plug)`	**WARN**
2.4	ChangeConfiguration AutoCharge=True → Accepted	`HTTP 200`	**PASS**
2.5	StatusNotification Preparing (AutoCharge=True)	`EV_WAIT_SEC=0 | skipped` *Connect physical EV to connector and rerun*	**WARN**
2.6	Authorize (VID) → Accepted	`Depends on 2.5 (EV plug)`	**WARN**
2.7	StartTransaction (session started)	`Depends on 2.5 (EV plug)`	**WARN**
2.8	StatusNotification Charging	`Depends on 2.5 (EV plug)`	**WARN**
2.9	MeterValues (measurands during charging)	`Depends on 2.5 (EV plug / active charging)`	**WARN**
2.10	RemoteStopTransaction → Accepted	`HTTP 200`	**PASS**
2.11	StopTransaction (sent after RemoteStop)	`session_state idle/finished not observed on MQTT in 30 s`	**WARN**
2.12	StatusNotification Finishing	`Finishing status not observed on MQTT` *Some implementations go directly Available without Finishing*	**WARN**

Item	Test Description	Detail / Evidence	Result
2.13	StatusNotification Available (after RemoteStop/unplug)	`Available status not observed on MQTT in 30 s`	WARN
2.14	StatusNotification Preparing (Session 2)	`EV_WAIT_SEC=0 | skipped` *Connect physical EV to connector and rerun*	WARN
2.15	Authorize (Session 2) → Accepted	`Depends on 2.14 (EV plug)`	WARN
2.16	StartTransaction (Session 2)	`Depends on 2.14 (EV plug)`	WARN
2.17	StatusNotification Charging (Session 2)	`Depends on 2.14 (EV plug)`	WARN
2.18	MeterValues (Session 2)	`Depends on 2.14 (EV plug)`	WARN
2.19	StatusNotification SuspendedEV	`Depends on 2.14 (EV plug)`	WARN
2.20	StopTransaction (reason=EVDisconnected)	`Depends on 2.14 (EV plug)`	WARN
2.21	StatusNotification Available (Session 2 end)	`Depends on 2.14 (EV plug)`	WARN

3 Notes on Failures and Warnings

FAIL items

- **2.1, 2.4** ChangeConfiguration AutoCharge: The `AutoCharge` key is not a standard OCPP 1.6 configuration key. The vSECC maps it internally to its OCPP 2.x variable `tx_ctrlr_tx_before_accepted_enabled`. The MEA CSMS may return `Rejected` for non-standard keys. If the CSMS returns `Accepted`, the item passes.

WARN items

- **2.2, 2.5, 2.14** StatusNotification Preparing: Requires a physical EV to be plugged into the vSECC connector. The test waits up to the configured `EV_WAIT_SEC` timeout. If no EV was present during the test run, these are recorded as WARN.
- **2.6–2.9, 2.11–2.13, 2.15–2.21** All items that depend on an active charging session (Authorize, StartTransaction, Charging, MeterValues, StopTransaction, etc.) are WARN when no EV is connected.
- **2.9, 2.18** MeterValues: The MEA sandbox REST API does not expose TriggerMessage (HTTP 404). MeterValues are observable only if the vSECC publishes them autonomously to MQTT based on `MeterValueSampleInterval`.
- **2.19** StatusNotification SuspendedEV: Requires the EV to stop drawing current (BMS charge limit reached or EV-side power management). Cannot be reliably triggered without a real EV battery management system.
- **2.20** StopTransaction (reason=EVDisconnected): Requires the user to physically unplug the EV connector.

4 Dependency Map

Items in Section 2 form two sequential charging session chains:

Session 1 2.2 → 2.3 (unplug)
2.5 → 2.6 → 2.7 → 2.8 → 2.9 → 2.10 (RemoteStop)
→ 2.11 → 2.12 → 2.13

Session 2 2.14 → 2.15 → 2.16 → 2.17 → 2.18 → 2.19
→ 2.20 (EVDisconnect) → 2.21

Items 2.1 and 2.4 (ChangeConfiguration) are independent of EV presence.